

Instructions

- There are 60 points on this quiz.
- You may work in groups of 2.
- You are not permitted to use your notes or computer.
- Show all work and clearly identify your final solution.

Name(s): _____

1. Let $f(y) = c \cdot y^2(2 - y)^2$ for $0 \leq y \leq 1$.

(a) (15 points) Find the value of c that makes this a valid pdf.

(b) (15 points) Find $E[Y]$.

2. Consider the Poisson distribution.

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, \dots$$

(a) (15 points) In class, it was shown that $E[X] = \lambda$. Show that $Var[X] = \lambda$.

(b) (15 points) Let X and Y be distributed as a joint Poisson with

$$f(x, y) = \frac{e^{-(\lambda_1 + \lambda_2)} \lambda_1^x \lambda_2^y}{x! y!}; \quad x = 0, 1, \dots; \quad y = 0, 1, \dots$$

If $X \sim Poisson(\lambda_1)$ and $Y \sim Poisson(\lambda_2)$, what is $Corr(X, Y)$?